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The Log of Gravity

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1. Introduction

Economists have long been aware that Jensen's inequality implies that $E(\ln y) \neq \ln E(y)$, that is, the expected value of the logarithm of a random variable is different from the logarithm of its expected value. This basic fact, however, has been neglected in many econometric applications. Indeed, one important implication of Jensen's inequality is that the standard practice of interpreting the parameters of log-linearized models estimated by ordinary least squares (OLS) as elasticities can be highly misleading in the presence of heteroskedasticity.

Although many authors have addressed the problem of obtaining consistent estimates of the conditional mean of the dependent variable when the model is estimated in the log-linear form (see, for example, Goldberger, 1968; Manning and Mullahy, 2001), we were unable to find any reference in the literature to the potential bias of the elasticities estimated using the log-linear model.

In this paper we use the gravity equation for trade as a particular illustration of how the bias arises and propose an appropriate estimator. We argue that the gravity equation, and, more generally, constant-elasticity models, should be estimated in their multiplicative form and propose a simple pseudo-maximum likelihood (PML) estimation technique. Besides being consistent in the presence of heteroskedasticity, this method also provides a natural way to deal with zero values of the dependent variable.

Using Monte Carlo simulations, we compare the performance of our estimator with that of OLS (in the log-linear specification). The results are striking. In the presence of heteroskedasticity, estimates obtained using log-linearized models are severely biased, distorting the interpretation of the model. These biases might be critical for the comparative assessment of competing economic theories, as well as for the evaluation of the effects of different policies. In contrast, our method is robust to the different patterns of heteroskedasticity considered in the simulations.

We next use the proposed method to provide new estimates of the gravity equation in cross sectional data. Using standard tests, we show that heteroskedasticity is indeed a